

Day 13 Activity: Reshaping data

Thursday, June 4, 2026

Your Name Here

Setup

```
library(tidyverse)
library(palmerpenguins)
```

Today's verbs are `pivot_longer()` and `pivot_wider()`. We'll use a few small built-in tables plus penguins.

Part 1

Look at `table1` and `table2` (both built into `tidyr`):

```
table1
```

```
# A tibble: 6 x 4
  country    year cases population
  <chr>      <dbl> <dbl>      <dbl>
1 Afghanistan 1999     745  19987071
2 Afghanistan 2000    2666  20595360
3 Brazil      1999   37737  172006362
4 Brazil      2000   80488  174504898
5 China       1999  212258  1272915272
6 China       2000  213766  1280428583
```

```
table2
```

```
# A tibble: 12 x 4
  country      year type      count
  <chr>      <dbl> <chr>      <dbl>
1 Afghanistan  1999 cases         745
2 Afghanistan  1999 population 19987071
3 Afghanistan  2000 cases        2666
4 Afghanistan  2000 population 20595360
5 Brazil       1999 cases        37737
6 Brazil       1999 population 172006362
7 Brazil       2000 cases        80488
8 Brazil       2000 population 174504898
9 China        1999 cases        212258
10 China       1999 population 1272915272
11 China       2000 cases        213766
12 China       2000 population 1280428583
```

1a. For each, describe what **one row** represents.

Table 1: [Answer here.]

Table 2: [Answer here.]

1b. Which one is **tidy** (each variable a column, each observation a row)? Which is the **narrower** of the two, and how can you tell?

[Answer here.]

Part 2

Build this wide table by hand:

```
bp_wide <- tribble(
  ~subject, ~before, ~after,
  "BHO", 160, 160,
  "GWB", 115, 135,
  "WJC", 105, 145
)
bp_wide
```

```
# A tibble: 3 x 3
  subject before after
  <chr>     <dbl> <dbl>
1 BHO         160   160
2 GWB         115   135
3 WJC         105   145
```

2a. Predict first. If you pivot `before` and `after` into a long form, how many **rows** will the result have? How many **columns**?

Prediction:

[Pred. # Rows here]

[Pred. # Cols here]

2b. Now pivot `bp_wide` to long form. Use `names_to = "when"` and `values_to = "sbp"`. Check against your prediction.

```
# Your code:
```

Part 3

The `billboard` data (in `tidyr`) has one column per chart week (`wk1`, `wk2`, ...).

3a. Pivot all the `wk` columns into `week` / `rank`, dropping the empty weeks. (Hint: `cols = starts_with("wk")`, `values_drop_na = TRUE`.)

```
# Your code:
```

3b. Add `mutate(week = parse_number(week))` so `"wk3"` becomes 3, then plot `rank` (y) against `week` (x), one faint line per track. Use `scale_y_reverse()` so rank 1 sits on top.

```
# Your code:
```

One sentence: what does the plot show about how long songs stay on the charts?

[Answer here.]

Part 4

Start from a long summary table:

```
penguin_summary <- penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(avg_mass = mean(body_mass_g, na.rm = TRUE), .groups = "drop")
penguin_summary
```

```
# A tibble: 6 x 3
  species sex    avg_mass
  <fct>   <fct>    <dbl>
1 Adelie female   3369.
2 Adelie male    4043.
3 Chinstrap female 3527.
4 Chinstrap male   3939.
5 Gentoo female   4680.
6 Gentoo male    5485.
```

4a. If you widen this with `names_from = sex`, `values_from = avg_mass`, how many rows and columns will you get?

Prediction:

[Pred. # Rows here]

[Pred. # Cols here]

4b. Pivot, then add a column `diff = male - female`.

```
# Your code:
```

Which species has the biggest mass gap between sexes?

[Answer here.]

Part 5

5a. Take your widened penguin table from Part 4 (before the `diff` column) and pivot it **back** to long form. Do you recover the original `penguin_summary`? (`pivot_longer()` and `pivot_wider()` are inverses.)

```
# Your code:
```

5b. Someone tries to make `bp_wide` longer like this and gets a strange result:

```
bp_wide |>
  pivot_longer(
    cols = c(subject, before, after),
    names_to = "when",
    values_to = "sbp"
  )
```

Run it. Why is this “too narrow” / wrong? What single change to `cols` fixes it?

[Answer here.]

Part 6 — Choose the shape

For each goal, say whether you’d want the data **wide** or **narrow**, and which pivot gets you there.

1. Plot each country’s TB cases over time, one line per country.
2. Compute, for each country, the change in cases from 1999 to 2000 as a single new column.
3. Group by year and sum the cases across all countries.

1. [Answer here.]

2. [Answer here.]

3. [Answer here.]

Code appendix

```
library(tidyverse)
library(palmerpenguins)
table1
table2
bp_wide <- tribble(
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  "GWB", 115, 135,
  "WJC", 105, 145
)
bp_wide
# Your code:

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penguin_summary <- penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(avg_mass = mean(body_mass_g, na.rm = TRUE), .groups = "drop")
penguin_summary
# Your code:

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bp_wide |>
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    cols = c(subject, before, after),
    names_to = "when",
    values_to = "sbp"
  )
```